

Timothy Singh

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Data Analyst

WORK EXPERIENCE

ALS Geoanalytics • North Vancouver, British Columbia, Canada

Apr 2025 - Jul 2025

Data Science Intern - Capstone Project

- Designed, developed and benchmarked a Python package to measure and track carbon emissions of large-scale deep learning models, achieving 85% measurement accuracy compared to existing packages.
- Integrated the framework with any Python workflow, enabling seamless adoption of this package to any PyTorch workflow.
- Leveraged real-time carbon intensity APIs to provide hourly regional emissions updates, ensuring 100% up-to-date tracking.
- Enabled tracking of carbon emission reduction techniques that reduced the carbon footprint of model training by up to 20%, using pruning and quantization, supporting company sustainability initiatives.

Ministry of Digital Transformation • Port of Spain, Trinidad and Tobago

Sep 2021 - Aug 2024

Data & IT Analyst

- Optimized inter-Ministry request and approval processes by developing a centralized online platform, reducing request processing time by 30% and increasing request visibility across 8 ministries.
- Engineered and deployed ICT Access Centres in 8 communities, providing underserved areas with access to cyber cafes and training rooms, impacting over 1,500 residents.
- Configured, deployed, and maintained Ubuntu 20.04 environments for 200+ staff using Bash scripting and network imaging tools, while executing bug tracking and unit-level testing to ensure reliability, resulting in a 40% decrease in on-boarding times.
- Implemented an open-source Asset Management system with LDAP-AD integration, automating asset tracking and improving deployment efficiency by 35%.
- Supported data cleaning and wrangling of budget spreadsheets to assist in fund allocation for fiscal 2023.
- Developed the Ministry's first global website, increasing domestic and international exposure by 90%.
- Managed and resolved Tier 0–3 technical support requests using ticketing software and remote troubleshooting tools (AnyDesk), handling 100+ issues monthly and independently resolving 85%.

Republic Bank Limited • Trincity, Trinidad and Tobago

Jun 2020 - Aug 2020

Data Analyst • Internship

- Cleaned and transformed large, complex Excel datasets using pandas (Python) to prepare structured inputs for ingestion into an internal SQL database, improving data accessibility for the analytics team.
- Designed and built a secure, front-end user interface using Apache Netbeans that enabled credit card administrators to manage database entries (create, edit, delete) without backend access, reducing administrative overhead by 70%.

SKILLS

- **Programming:** Bash, C#, C++, Java, JavaScript, Python, SQL
- **Web & Backend:** CSS3, Express.js, Flask, HTML5, Node.js, REST APIs, Spring Boot
- **Machine Learning & AI:** CNNs, Ensemble Models, NLP (spaCy, NLTK, BERT), PyTorch, RNNs, scikit-learn, SHAP, Transformers
- **Data Analysis & Visualization:** Altair, Dash, ggplot2, Matplotlib, NumPy, Pandas, Power BI, seaborn, Shiny
- **Statistics:** A/B Testing, Hypothesis Testing, Regression, Time Series (ARIMA, LSTM)
- **Cloud & DevOps:** AWS (S3, Lambda, EC2), Azure, CI/CD (GitHub Actions), Docker, Git, Linux
- **Big Data & Databases:** Hadoop, MapReduce, Spark, SQL/NoSQL (Oracle, MySQL, PostgreSQL, MongoDB)
- **Tools:** Eclipse, IntelliJ, Jupyter, NetBeans, VS Code

EDUCATION

M.Sc. in Data Science

University of British Columbia • Vancouver, British Columbia, Canada • GPA: 95.0

Aug 2024 - Jul 2025

B.Sc. in Computer Science

University of the West Indies • St. Augustine Campus, Trinidad and Tobago • GPA: 4.23

Sep 2018 - Jul 2021

CERTIFICATIONS

Generative AI: Prompt Engineering Basics

Jun 2024

Create Machine Learning Models in Microsoft Azure

Mar 2024

Introduction to Artificial Intelligence

Jan 2024

Generative AI: Introduction and Applications

Jan 2024

Microsoft Office Specialist – Microsoft Excel Certification

Jan 2021

PROJECTS

dsresumatch

<https://github.com/UBC-MDS/dsresumatch>

DSResuMatch is a package designed to analyze resumes and job descriptions by extracting key sections and scoring them based on keyword relevance.

Credit Card Default Predictor

<https://github.com/SimplyTim/Credit-Card-Default-Predictor>

This project aims to predict the likelihood of a customer defaulting on their credit card payments using historical payment data. The model incorporates various feature engineering techniques to improve predictive accuracy and reduce overfitting. The dataset was obtained from Kaggle containing around 30000 rows with 25 features.

NIH X-Ray Predictor

<https://github.com/SimplyTim/NIH-Chest-XRay-Classifer>

This project is part of the 2025 UBC Medicine Datathon, where we attempt to predict and classify diseases given X-ray images from the NIH Chest X-rays Dataset on Kaggle.